

Cross-Architecture GPU Fault Injection at the LLVM IR Level

Maisy Dunlavy¹, Zhiling Lan^{1, 2}, Michael Papka^{1, 2}

¹University of Illinois Chicago, Chicago, IL, USA — ²Argonne National Laboratory, Lemont, IL, USA

Motivation and Gap

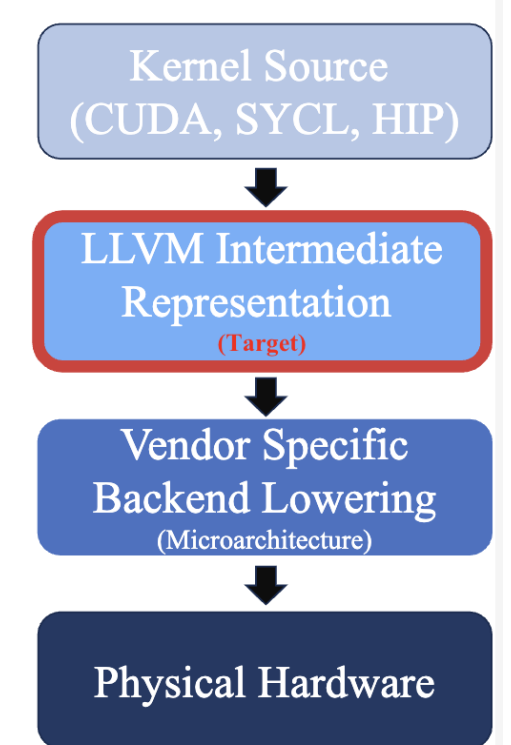
- GPU resilience studies are usually performed on **one vendor platform at a time**, using a vendor-specific fault injection tool.
- Impossible to know whether conclusions about masking, corruption, or failure can generalize across most common GPU platforms used in HPC.
- Backend-specific binaries differ substantially; if faults are injected only after vendor-specific lowering, they are no longer directly comparable.

Prior to this work, there was no way to study the outcome of the same logical fault across NVIDIA, AMD, and Intel GPUs.**

**As far as we are aware.

Our Approach

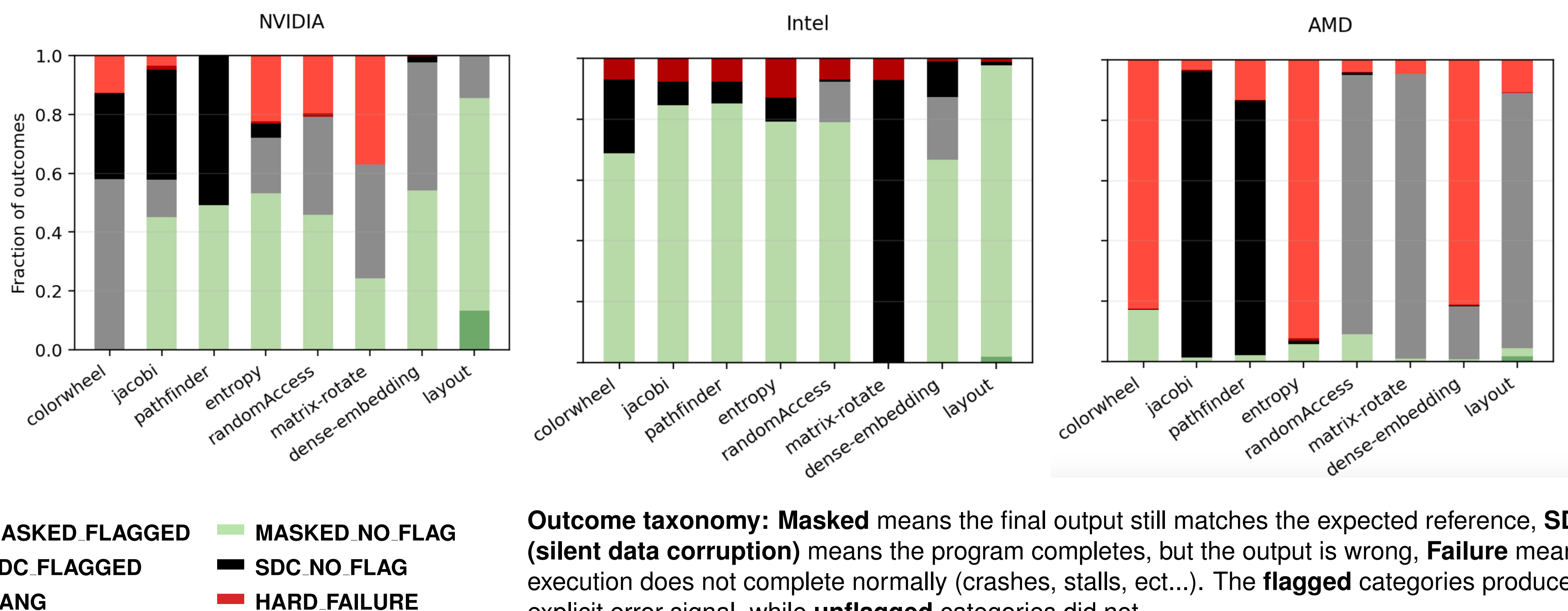
- **Faults are injected before vendor-specific lowering**, using a shared intermediate program representation: LLVM IR.
- **Same logical fault** is carried through each platform's native compilation flow and program stack.
- **Directly equivalent** cross-platform comparison across GPU backends.
- **Observed differences** can be tied to **GPU backend** and **workload characteristics**.



Results

Key result: Resilience is **not backend-invariant**. Under the same logical fault model, NVIDIA, Intel, and AMD backends show different mixtures of masked outcomes, silent corruption, and failures. Single-vendor resilience results should therefore not be assumed to transfer across GPU platforms.

All experimentation was conducted using the HeCBench Benchmark Suite [1] on Polaris [2], Aurora [3], and Frontier [4].



Outcome taxonomy: **Masked** means the final output still matches the expected reference, **SDC (silent data corruption)** means the program completes, but the output is wrong, **Failure** means execution does not complete normally (crashes, stalls, ect...). The **flagged** categories produced an explicit error signal, while **unflagged** categories did not.

Conclusions

- Resilience behavior is **not portable** across GPU vendors.
- A single-platform fault study may miss important backend-specific effects.
- Successful execution does **not** always imply correct execution.
- LLVM IR-level injection enables more directly comparable cross-platform analysis.

Moving Forward

These results motivate:

- broader cross-architecture resilience studies,
- backend-aware mitigation strategies,
- workload-aware protection,
- more rigorous evaluation methods for heterogeneous HPC systems.

References

References

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- [5] LLVM Project. LLVM 17 release. <https://releases.llvm.org/>, 2023. Accessed: 2026-04-29.

Contact and Funding

For questions, collaboration, or follow-up materials, scan the QR code or contact:

Maisy Dunlavy - mdunl6@uic.edu

This work is supported by the Chicagoland Computational Traineeship in High Energy Particle Physics, which is in partnership with Argonne National Laboratory and Fermi National Accelerator Laboratory.

